

Process Guidelines to Ensure Optimal SMT Electronics Assembly

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Abstract

The continuing miniaturization of personal electronics devices, such as mobile phones, personal music devices, and personal computing devices has driven the need for increasingly smaller active and passive electrical components. Not too long ago, 0402 (40 x 20mils) passives were seen as the ultimate in miniaturization, but recently 0201 and now 01005 passives have arrived, with predictions of even smaller sizes to come. For active components, the 30mil CSP (a chip-scale package with the solder balls on 30mil/0.75mm centers) has virtually become a requirement for enabling the many features required in modern portable electronics devices. This miniaturization trend, occurring simultaneously with the conversion to RoHS-compliant lead-free assembly, has put a considerable strain on the electronics assembly industry.

This paper will discuss guidelines to optimize the electronics assembly process from stencil printing to reflow profiling. With a smaller process window, each variable in the process contributes to the overall success of the assembly. Simple contributions such as storage and handling of solder paste can make or break a printing process. Many solder defects, some say 60-70%, can be attributed to the stencil printing process. Therefore, stencil printing setup and solder paste measurement metrics are discussed to determine the potential for success. Other topics include: stencil guidelines, solder paste technology with reflow profile considerations, and proper thermocouple attachment. Acting on all of this information will help ensure a reliable solder joint.

Key Words: Electronics assembly, stencil printing, solder paste, process optimization, reflow profiling, thermocouples, ramp-to-peak, graping, head-in-pillow

Introduction

The challenge to ensure good solder joint integrity has been impacted by the higher reflow temperatures required by lead-free solders. Perhaps more importantly, the continual move towards miniaturization has compounded the issue as stencil printing, component placement, and

reflow soldering of these ultrafine components require precision, accuracy, and a pristine process. Newer assembly technologies that have been introduced to help meet these challenges will also be discussed. However, some fundamental practices once easily overlooked, such as the importance of storage and handling

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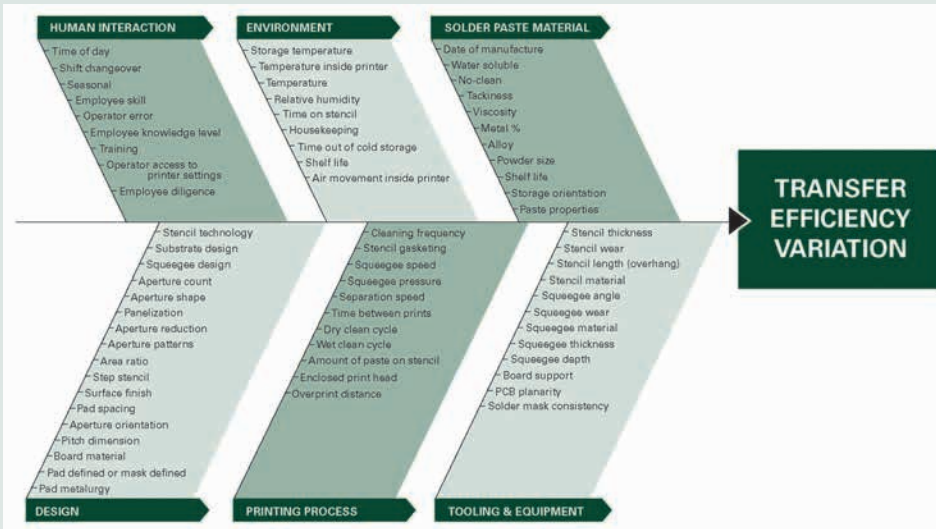


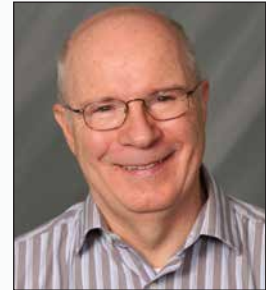
Figure 1. Stencil printing Ishikawa diagram (larger image can be viewed at the end of this paper).

of solder paste, need to be re-visited. These basic practices may be forgotten or overlooked and need to be confirmed once again.

It is commonly accepted that 60-70% of solder defects occur at the printing stage. The Ishikawa diagram in Figure 1 shows the significant number of issues that can present themselves in the stencil printing process. This diagram can help one to understand why it can be the root of many solder defects. With

the continuation towards component miniaturization, the stencil printing process window becomes even more challenging, increasing the opportunity for insufficient or inconsistent solder paste deposits. The “math” is easy—no or too little solder paste deposited onto the PWB pad results in no solder joint. Too much solder paste can cause an electrical short circuit. Many stencil printer platforms offer inspection equipment to catch such discrepancies.

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Paste Measurement

The most important tool to measure the efficiency of the stencil printing process is a solder paste inspection (SPI) machine. SPI equipment can be utilized to detect insufficient or excessive solder paste deposits and to determine if the solder paste deposit is located squarely on the PWB solder pad.

Hence, SPI enables accurate decisions about the volume and position of the solder paste deposits. Such data can eliminate subjective and perhaps erroneous decisions from the decision making process. The SPI data can also alert the process engineer to stencil printing issues that otherwise may have been overlooked.

Just as important for ultrafine pitch components is pad-to-pad solder paste deposit consistency for a particular component. This consistency may mean

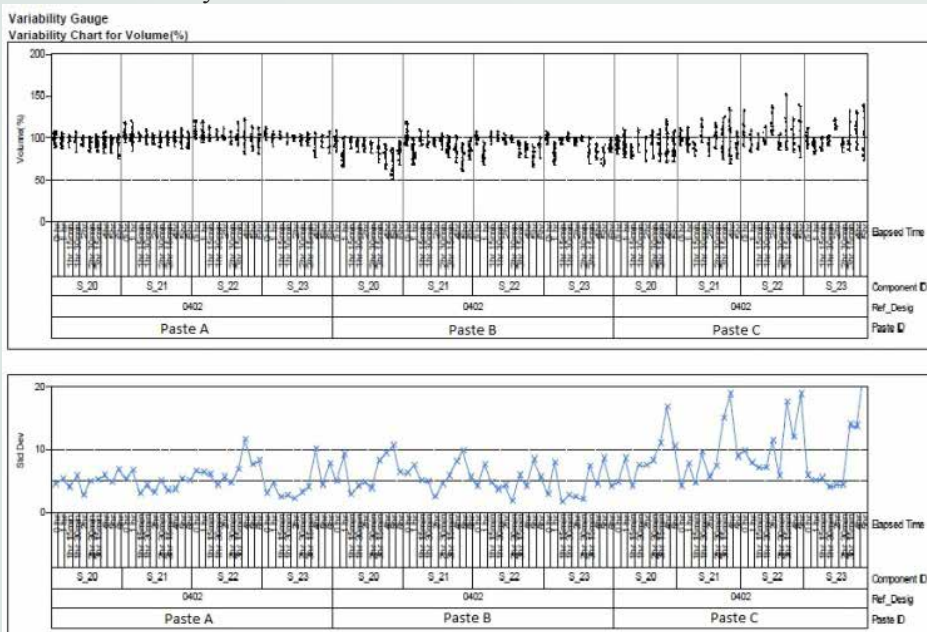


Figure 2. Transfer efficiency and standard deviation for 0402 print study using Paste A, B, and C.

the difference between a discrete component tombstoning or, especially for components such as ball grid array (BGA), head-in-pillow defect. Looking at Figure 2, one may assume that solder paste C, providing the greater average volume would be the preferred solder paste. However, looking at the consistency/standard deviation chart, below the box plot diagram, paste A and B, though with a bit lower volume, are more consistent. Consistency in stencil printing is the most important metric. Looking at two examples will explain why this is the case. Example 1 is a discrete component; tombstoning occurs when there is a force imbalance at either end of the component. There are many reasons this force imbalance may occur, including instances where more solder paste is deposited on one pad than the other. This situation creates an unequal pull on one end of the component and can result in a tombstone.

Example 2 is a BGA-type component. Consider the situation where there are three consecutive pads in a row. If the two outer pads have significantly more solder paste deposited on them than the pad in the center, the potential for the head-in-pillow defect will significantly increase for the center connection.

As mentioned before, for ultrafine solder paste deposits, paste volume measurement is vital in ensuring consistency as well as the average volume. Other methods are subjective, determining only that this is or is not paste on the pad. Older SPI technologies may not have the precision needed for smaller features. These older units often have a Gage R&R Precision Tolerance Ratio (PTR) of 25-30% for the smallest solder paste deposits; whereas, a PTR of less than 10% is recommended. Most new SPI machines have a PTR of less than 5% (Figure 3).

When discussing solder paste measurement, there are two important metrics to consider in the printing process: the area ratio and transfer efficiency.

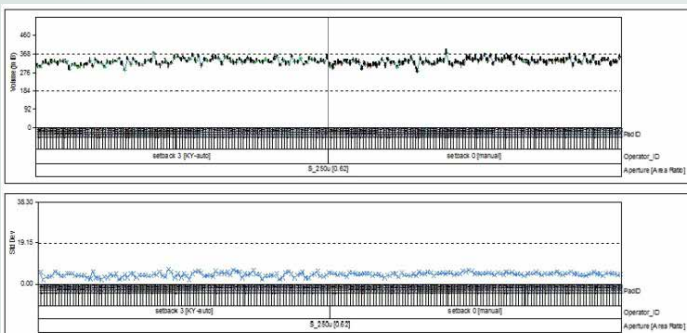
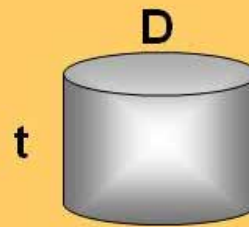


Figure 3. Typical Gage R&R results showing less than 5%.

Area Ratio



$$A_{\text{opening}} = \pi D^2/4$$

$$A_{\text{wall}} = \pi D t \Rightarrow$$

$$AR = D/4t$$

Figure 4. A schematic showing the definition of the area ratio for a circular stencil.

The area ratio (AR) is a critical metric for successful stencil printing; it is the area of the stencil aperture opening divided by the area of the aperture side walls. Figure 4 shows a schematic for a circular aperture. A simple calculation shows that the area ratio (AR) is the diameter (D) of the circle divided by 4 times the stencil thickness (t): $AR = D/4t$. Somewhat surprisingly, the result is the same for square apertures, with D now equal to the side of the square. For the AR of a rectangular aperture, the formula is a little more complicated: $ab/2(a+b)t$, where a and b are the sides of the rectangle.

It is widely accepted that in order to get good stencil printing, the AR must be greater than 0.66. Experience has shown that if $AR < 0.66$, the transfer efficiency will be low and erratic. Transfer efficiency, another important stencil printing metric, is defined as the volume of the solder paste deposit divided by the volume of the aperture.[1]

Solder Paste Storage and Handling

Often overlooked is the importance of the storage and handling of the solder paste. Solder paste is a perishable material and an interesting and comparable analogy can be made with milk, also a perishable product. Both have a shelf life and even though refrigerated, both will “spoil” beyond the expiration date. The solder paste flux is always “working” even when refrigerated, removing surface oxides from the solder powder particles within the solder paste itself. The reaction is much slower when refrigerated, but occurs nonetheless. The byproduct of this reaction is a heavy metal salt resulting in an increase in the solder paste viscosity. The solder paste “roll” on the stencil is affected, hampering the ability of the solder paste to fill the stencil apertures, resulting in insufficient paste deposits on the PWB pads. The available flux when needed is also diminished as the flux activity is “consumed” in this chemical reaction; there is less flux activity to overcome the challenges of the reflow profile. With

diminished flux during reflow, the solder paste is even more challenged to overcome solder defects that are sensitive to oxidation, such as graping, head-in-pillow, and solder balls. This challenge is even more evident as the solder paste deposit becomes smaller and smaller.

Returning to our analogy of milk: once removed from the fridge, how long can I leave it on the counter before it spoils? This is a difficult question to answer with a one size fits all response. There are a number of variables to consider: how close is the milk to the expiration date (the closer it is to the expiration date, the faster it will spoil); how much milk is in the container (a full gallon will take longer to spoil than a gallon container with just a little milk in it); how long was the cap left off the container; how many times was the cap removed; did my son contaminate it by drinking out of the container (for solder paste this would be adding old paste back into the remainder of the fresh paste in the jar); temperature, and humidity, to name a few. Any combination of these causes the solder paste to “spoil” at an even greater rate.

I should mention that the analogy falls short in a couple respects; cycling milk in and out of the refrigerator helps slow the spoiling process for milk, but it actually accelerates the “spoiling” process for solder paste. Secondly, it is imperative that the solder paste be thawed to room temperature before use. If not thawed completely, moisture can condense on the solder paste much like evidenced on a gallon milk container removed from the refrigerator. This moisture also accelerates the “spoiling” process. Improper thawing, such as setting the solder paste on a window sill in the sunlight or on top of an oven, will also accelerate the spoiling process.

Flux Chemistry

The type of flux chemistry, specifically water-soluble vs. no-clean, also plays a significant role as to the sensitivity of the solder paste in the environment that it is exposed to. Water-soluble chemistries are by their nature much more sensitive to temperature/humidity as they are more hygroscopic than their no-clean rosin/resin based counterparts. The smaller paste deposits required to solder ultrafine pitch components increases this sensitivity.

Rosins/resins (hereafter referred to as resin) play a significant role in flux chemistry. Right out of the refrigerator, resins offer a longer shelf and stencil life, are more consistent in behavior through seasonal changes, have better print performance overall, have more tack to hold components in place, and provide better oxidation resistance to the solder powder during the reflow process.

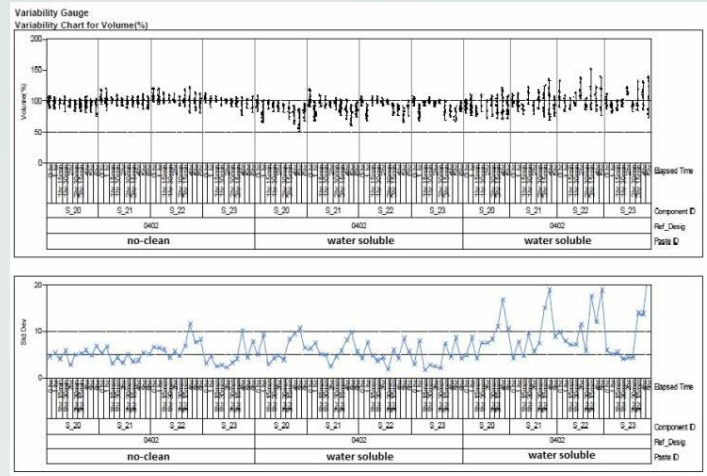


Figure 5. Example of the print performance of no-clean vs. water-soluble solder pastes.

Figure 5 represents typical results of comparison of no-clean vs. water-soluble print performance. Resins are tacky substances which increase adhesive (external attraction of the solder paste to other surfaces) and cohesive (internal attraction between the flux-coated solder particles within the solder paste itself) properties. The cohesion helps the solder paste release from the stencil aperture. In addition, the cohesive forces help the resulting solder paste deposit retain the shape of the stencil aperture and resist “slump.” As the solder paste is printed onto the PWB pad, the adhesion of the solder paste pulls the newly deposited solder paste from the stencil aperture as the PWB is lowered from the stencil. For ultrafine stencil printing, these forces are vital to ensuring success.

Water-soluble fluxes contain more activators than no-clean fluxes. Although water-soluble chemistries are more active initially in removing existing surface oxides, they can falter during the reflow process after the existing oxides are removed. Re-oxidation takes place later in the reflow profile

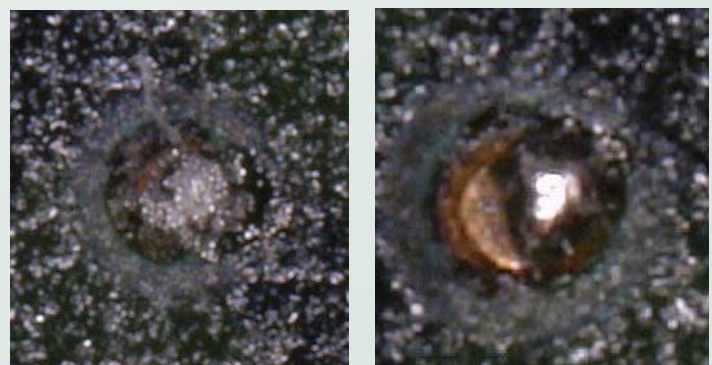


Figure 6. Typical results - water-soluble (left) vs. no-clean (right) using the same Type 6 powder size and reflow profile (RTP).

where higher temperatures accelerate the rate of oxidation and the water-soluble chemistries do not provide the oxidation barrier needed to prevent this from happening. Even though water-soluble chemistries are initially more active, no-clean chemistries can often provide better results from stencil printing through reflow.

Figure 6 represents a comparison of a water-soluble solder paste reflow result versus no-clean solder paste. The materials were Type 6 powder solder pastes, 6mil stencil apertures with a 3mil thick stencil, using a high temperature soak reflow profile. These factors represent a challenging reflow environment, but signify a common trend in comparing the no-clean and water-soluble solder pastes. The pictures show that the water-soluble solder paste produces what appears to be a cold solder joint when in actuality the activator was exhausted and with very little oxidation resistance the powder particles became oxidized and could not coalesce. The no-clean solder paste under the same exact reflow profile reflowed and coalesced fairly well.

TYPE Per J-STD-005	Aperture Width (Avg.)				Diameter Range			
	millimeters	inches	# small spheres	# large spheres	microns		inches	
3	0.24	0.0095	9.7	5.4	25	45	0.0010	0.0018
4	0.24	0.0095	12.1	6.4	20	38	0.0008	0.0015
3	0.19	0.0075	7.6	4.2	25	45	0.0010	0.0018
4	0.19	0.0075	9.5	5.0	20	38	0.0008	0.0015
4	0.14	0.0055	7.0	3.7	20	38	0.0008	0.0015
5	0.14	0.0055	9.9	5.6	15	25	0.0006	0.0010
6	0.10	0.004	20.3	6.8	5	15	0.0002	0.0006

Figure 7. Aperture width vs. number of large spheres.

Figure 7 gives a general recommendation when to consider the use of finer powder particle size solder pastes for a particular printing application. The general rule of thumb is that a minimum of 4-6 of the largest particles within the solder paste laid side-by-side should fit within the width of the smallest aperture on the stencil.

As seen in Figure 8, the finer particle size (Type 5) exhibits less scatter (fewer outliers) in the uppermost chart showing transfer efficiency and a decrease in the standard deviation (lower graph) meaning improved stencil printing consistency.

However, as finer solder powders are utilized in manufacturing the solder paste, the total surface area of the solder powder in the solder paste deposited increases. Increased surface area also means there is an increase in the amount of surface oxide present. This surface oxide further challenges the capability of the flux to remove all the existing surface oxides and to protect from further oxidation during reflow. For smaller print deposits required for ultrafine stencil printing, small solder

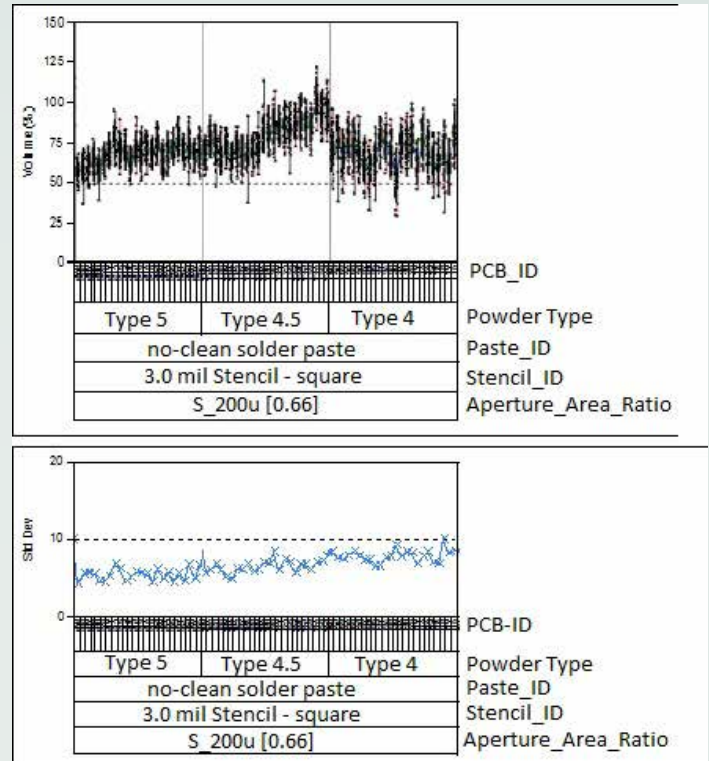


Figure 8. Particle size comparison, same no-clean flux formula.

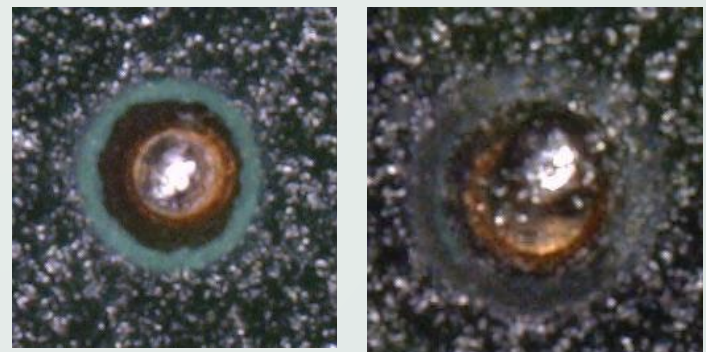


Figure 9. Type 3 (left) vs. Type 6 (right), same no-clean flux formula and reflow profile (RTP).

paste deposits may not have enough flux to provide enough activity or oxidation resistance, making it more prone to solder defects associated with the reflow profile (Figure 9).

Printer

Proper maintenance and good housekeeping of the stencil printer should not be overlooked. Solder paste or other debris on the stencil printer conveyor belts, lift table, board support system, or underside of the stencil can act as a shim, therefore preventing the PWB from gasketing to the underside of the stencil. The lack of gasketing will no longer confine the solder paste to the PWB pad, but will allow solder paste to flow under the stencil to unwanted areas.

Proper housekeeping also extends to the squeegee blades and peripheral utensils used in the stencil printing process. Utensils, such as putty knives and jars used to store solder paste removed from stencils, are also important to keep clean. Dry solder paste from the threads of a jar or from unclean utensils can fall into the solder paste transferred to the stencil. The dried paste will inhibit the paste from filling the stencil apertures or will clog the stencil apertures. Squeegee blades, utensils, and jars used to store paste from the stencil at the end of the day should be clean.

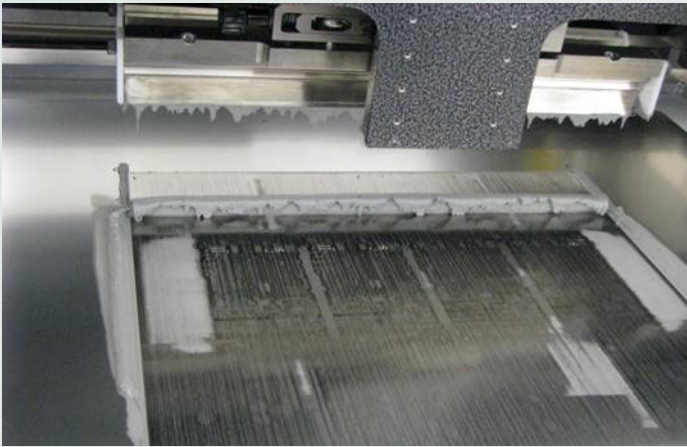


Figure 10. Squeegee blades that are too long.

The most important aspect of the printer setup for fine-feature stencil printing is stencil-to-PWB gasketing and registration. As mentioned above, these factors are critical in ensuring that solder paste is confined only to the PWB pad. With the PWB in position and mated to the underside of the stencil, press down on the stencil in several locations; there should be no movement of the stencil. One area that can help ensure good gasketing is through board support. Experimentation has confirmed that a solid support with landscape vacuum tooling provides the best board support and gasketing.

Along with regular preventative maintenance, equipment calibration ensures that the table, rails, conveyor, etc. are parallel, aiding good registration and gasketing.

Sample Area Ratio Chart												
Pad Size (mm)	0.85	0.10	0.15	0.1005	0.20	0.25	0.201	0.30	0.35	0.40	0.45	0.50
Aperture Width (mil)	2.0	3.9	5.9	7 x 8	7.9	9.8	10 x 12	11.8	13.8	15.7	17.7	19.7
Stencil Thickness (5.0 mil)	0.10	0.20	0.30	0.37	0.39	0.49	0.55	0.59	0.69	0.79	0.89	0.99
Stencil Thickness (4.5 mil)	0.11	0.22	0.33	0.41	0.44	0.55	0.61	0.66	0.77	0.87	0.98	1.09
Stencil Thickness (4.0 mil)	0.12	0.25	0.37	0.47	0.49	0.62	0.68	0.74	0.86	0.98	1.11	1.23
Stencil Thickness (3.5 mil)	0.14	0.28	0.42	0.53	0.56	0.70	0.78	0.84	0.98	1.12	1.27	1.41
Stencil Thickness (3.0 mil)	0.16	0.33	0.49	0.62	0.66	0.82	0.91	0.98	1.15	1.31	1.48	1.64
Stencil Thickness (2.5 mil)	0.20	0.39	0.59	0.75	0.79	0.98	1.08	1.18	1.38	1.57	1.77	1.97

Figure 11. Stencil thickness vs. area ratio (dark green - poor area ratio, light green - borderline area ratio, medium green - good area ratio) (larger image can be viewed at the end of this paper).

As seen in Figure 10, squeegee blades that are much longer than the board width can leave thin trails of solder paste if the board at the outer edges of the squeegee blades is not supported. This thin layer of solder paste will dry out and oxidize very quickly, and if added back to the solder paste bead, will inhibit the solder paste from filling the stencil aperture.

Stencil design should follow the previously mentioned area ratio metric when possible. A simple way to increase the area ratio is to reduce the stencil thickness (Figure 11). However, care needs to be taken with smaller apertures that by reducing the stencil thickness, there will still be sufficient solder volume for larger components.

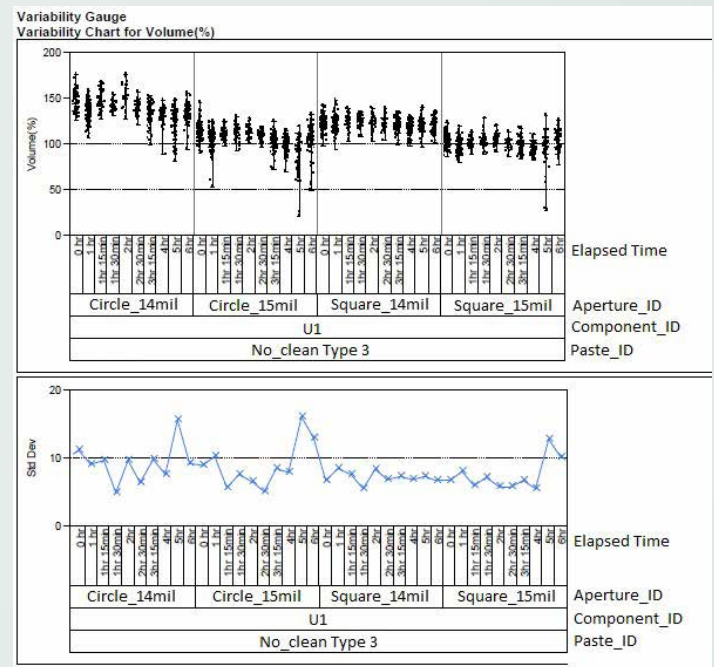


Figure 12. Round vs. square aperture comparison.

Figure 12 shows that stencil printing for square (with radius corners) apertures is more consistent than circular apertures of the same area ratio. The paste releases more consistently with the square apertures. Part of the reason for this better consistency is that for the same area ratio, the square provides more volume than the circular aperture.

Figure 13 shows that stencil printing for smaller apertures is aided by the use of solder mask defined pads. Note that there is significantly less scatter for both 8mil circles and squares that are solder mask defined. The solder mask defined pads exhibit more consistency in the volume of the solder paste deposit. Paste release from the stencil aperture is aided by the adhesion of the solder paste to the PWB pad. The adhesion

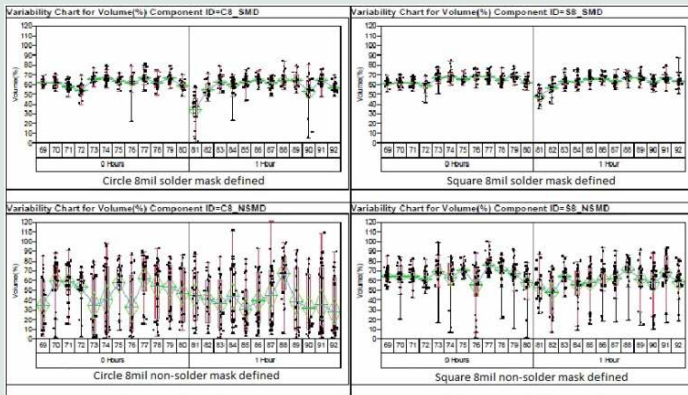


Figure 13. Top row: solder mask defined 8mil circles and squares respectively. Bottom row: non-solder mask defined (or pad defined) 8mil circles and squares respectively.

helps to pull the solder paste from the stencil aperture as the PWB is lowered from the stencil. For smaller apertures, there is very little surface area for this adhesion to occur. The walls of the solder mask itself add to the total amount of surface area for the solder paste to adhere to. The increased surface area aids in pulling the solder paste from the stencil aperture.

Perhaps one of the more exciting technologies to enhance the stencil printing process is the introduction of nano-coatings. The use of these coatings, more refined lasers, and finer grain stencil materials create a cleaner cut of the stencil aperture walls, allowing for better release of the solder paste from the stencil aperture. Almost to the extent of the introduction of Teflon® coatings in pans to the cooking industry, these nano-coatings have similarly improved the performance of stencil printing. The introduction of these technologies may have improved the solder paste release from the stencil to the point that the suggested guidelines of an area ratio of 0.66 may be relaxed in the near future.

Summary of Solder Paste Technology

In today's technology, the higher reflow temperatures associated with the use of lead-free assembly processes and the continual move towards miniaturization have caused a closer look at existing practices in stencil printing and reflow profiling. Even the practice of storage and handling of solder paste, general housekeeping of the stencil printer, and the peripheral utensils utilized with it, need to be re-visited to ensure success. It is well-accepted that 60-70% of solder defects occur at the stencil printer. Metrics such as area ratio and transfer efficiency play a significant role in stencil printing consistency/success. Paste measurement equipment provides

an excellent tool in understanding, and perhaps overlooked, deficiencies in a stencil printing process. Paste measurement provides the tool for making decisions to statistically make change.

Reflow Profiling

Why Profile?

It is well known that the typical lead free SAC (SnAgCu) alloys require higher process temperatures during reflow than tin-lead solders. The FR4 laminate used in PWB manufacturing, the moisture sensitive polymers used to encapsulate IC components, and heat sensitive components such as LEDs are more prone to heat induced damage due to the higher temperature requirements for lead free assembly. However, the continuation towards miniaturization has significantly added to the challenge of successful reflow profiling. The small print deposits required for ultrafine stencil printing are very sensitive to the environment to which they are subjected and a reflow profile that may have worked very well in the past may exhibit solder defects for these smaller solder paste deposits. It is important to note that this sensitivity is not only in the reflow oven, but begins right at the stencil printer after the solder paste is deposited onto the PWB. These smaller solder paste print deposits have little flux reserve to prevent the solder paste from drying out and from oxidation, thus decreasing the success of forming a good solder joint in the reflow process. Common issues for these smaller solder paste print deposits are graping and the head-in-pillow defect. Documenting the thermal profile of the reflow oven for each electronics assembled product is essential for analyzing solder defects and creating an audit trail. The reflow profile is typically the first thing a process engineer should examine in troubleshooting an assembly line, if it is producing too many solder defects.

TC Types

Thermocouples (TC) are comprised of two dissimilar metals joined together at the tip with a welded bead. The two metals produce a voltage when subjected to a differential in temperature. A number of thermocouple types are available, generally Type K is selected for SMT assembly. Type K thermocouples are comprised of a nickel-chromium positive lead and a nickel-aluminum negative lead with an operating range of -200 - 1250°C ± 1.5°C.

TC Attachment Methods ^{2,3}

(The following information appears courtesy of KIC)



Figure 14. Thermocouples should not be cut or twisted.² (Courtesy of KIC)

The voltage produced is measured where the two dissimilar metals first contact each other. Ideally measuring at the bead at the tip of the wires; however, if the wires are twisted or the insulation on the wires is worn through, the TCs will read at the first point the wires come together rather than at the tip where the TCs are attached, giving erroneous results. The quality of the attachment is perhaps even more important than the data logger itself, as the data logger cannot correct for erroneous readings initiated at twisted connections or wire contact due to worn insulation.

The material used to attach the TCs should provide good contact of the TC bead to the data point where the temperature is to be measured and should not interfere with the bead's ability to accurately record the temperature changes the data point is experiencing during reflow.



Figure 15. Profile using Kapton tape attachment. Note fluctuating data. (Courtesy of KIC)



Figure 16. Note that process window index indicates 139% due to loose and erroneous data. (Courtesy of KIC)

Epoxy does a good job of holding the TC in place and providing good contact, but the amount and type of epoxy is critical to successful data collection. If too much epoxy is added, the TC does not react to the changes in temperature quickly enough. If the epoxy is not thermally conductive, it acts as an insulator and can result in erroneous readings.

High temperature solder gives the best contact, but the amount used can also slow the response of the TC to changes in temperature.

Kapton tape offers ease of use, in securing thermocouple beads, but may not offer good contact. As seen in Figure 15, the bouncing data indicates that the Kapton loosened during the profile giving fluctuating or inaccurate data.

As seen in Figure 16, the loose connection portrays a process window index that is out of specification. This erroneous data also confuses the ability of the software to make an accurate prediction.

In this instance, after correcting the loose TC connection the process was actually within specification.

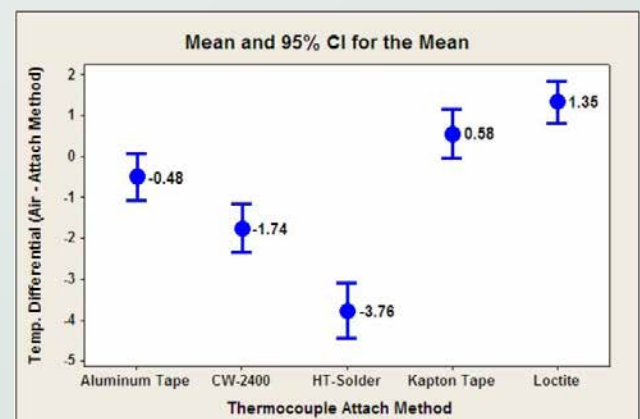


Figure 17. TC attachment method analysis. (Courtesy of KIC)

Aluminum tape offers more adhesion than Kapton tape, creating a better contact between the TC bead and the data point on the test board. In addition, aluminum tape is an excellent thermal conductor. Therefore, it provides good response to changes in temperature, giving accurate and timely readings. In reflow profile evaluations, aluminum tape produced the smallest standard deviation. Its adhesion can loosen over subsequent profiles, so if multiple profiles are required, Kapton tape can be used in conjunction with aluminum tape or the engineer can simply re-apply new aluminum tape.

Thermocouple Bead Attachment Location

Reflow profile measurement is best done on the actual fully assembled production board. It is the most accurate method available. A minimum of three thermocouples should be attached; one attached to the hottest location, another to the coolest, and the third attached so that it protrudes (~1 inch) from the leading edge to record air temperature within the oven during reflow. The thermocouple recording air temperature may or may not be required by the data logger itself, but the data collected is useful in providing feedback as to whether the oven is performing as expected. For instance, if the oven zone settings are 120/140/160/180/200/220/240 °C, one would expect the readings from this thermocouple to read each of these zone settings within that zone. Additional thermocouples can be used for temperature sensitive components, edge of PWB vs. middle, BGA, etc. as needed.

Ball grid array (BGA) packages offer a bit of a challenge for reflow profiling, as the soldered connection is beneath the package. For such TC connections, the most accurate data are collected by attaching a thermocouple to a sphere near the center of the package. This technique can be accomplished by drilling into a solder sphere (not through the solder sphere) on an assembled PCB or by pre-drilling the hole prior to stencil printing. The stencil will need a stencil wipe, as solder paste will protrude through the bottom side of the stencil, where the hole was drilled into the PWB. The assembly should then be reflowed after the thermocouple is attached through the drilled hole, ensuring it contacts the solder sphere. Adhesive can be used to hold the TC wire in place.

Profile Types

Ramp-to-Peak (RTP)

Not long ago the “soak profile” was the profile of choice. However, as mentioned previously, the introduction of higher

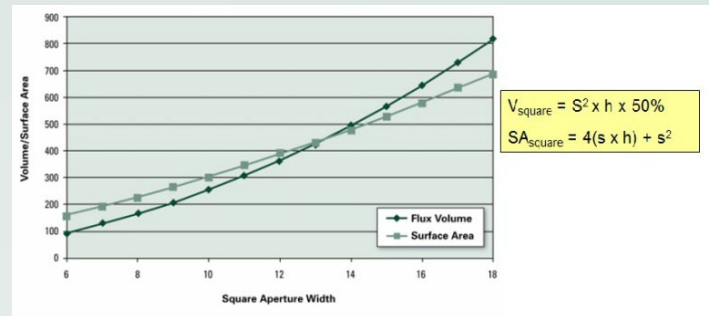


Figure 18. As the aperture width decreases from 18mil to 6mil, the ratio of exposed surface area to available flux increases.

reflow temperatures associated with lead-free assembly, and dealing with temperature sensitive components, profiles that reduce the total heat input have been adapted.

In addition, miniaturization has decreased the size of the solder paste print deposit required to solder smaller components such as 0201 and 01005. Figure 18 shows the relationship of increased surface area to available flux to remove surface oxides and prevent further oxidation, making the smaller deposits much more sensitive to the reflow profile.

RTP vs. Soak



Figure 19. Typical results - RTP profile (left), soak profile (right) using the same Type 6 powder size and flux chemistry (no-clean).

The combination of higher reflow temperatures and the sensitivity of the smaller solder paste print deposits has led to the establishment of the ramp-to-peak (RTP) or ramp-to-spike profile, as the new standard for reflow soldering. The RTP profile reduces the thermal stress and total heat input the assembly experiences during the reflow process. As seen in Figure 19, the reduced thermal input provided by the RTP can produce substantially improved results.

Max Slope vs. Ramp Rate

When discussing reflow profiling, the terms peak temperature and time-above-liquidus are critical, and these parameters are typically well-defined. Perhaps a little misunderstood, or

at least easily overlooked, is the importance of the preheat portion of the reflow profile. In lead-free small component assembly, this portion of the profile has become equally as important. The preheat portion of the profile is controlled by the belt or conveyor speed and the zone temperature settings in the initial zones of the oven. Typically, the last two zones of the reflow oven are where the peak temperature and time-above-liquidus are controlled.

Many solder paste technical data sheets define parameters for the ramp rate or the ambient-to-peak temperature divided by the time. Many process engineers assume this terminology is the same for what some data loggers refer to as max(+)slope. However, the terms are actually referring to parameters that are quite different. Max(+)slope is geared more to component manufacturer recommendations so as not to thermal shock components. Typically the component manufacturers prefer that the components not experience more than 3°C/second. The data point displayed for max(+)slope is merely a snapshot in time where the maximum change in temperature occurred. It is generally within the first zone, as the assembly enters the first zone at ambient temperature (25°C) and is subjected to the temperature in the first zone, which is typically about 100°C. This represents a 75°C delta. Typically the delta between the other zone temperatures is $\leq 40^\circ\text{C}$.

There is a balance, as with any part of the SMT process, between a ramp rate that is too fast and too slow. Too fast and the solder paste slump increases. Slump is the spreading of the solder paste that induces solder bridging resulting in solder shorts. In addition, because the solder paste solvent has not yet been driven off, flux explosions or solder splatter can cause solder balls and solder beads to occur. If the ramp rate is too slow, heat or thermal-related solder defects, such as graping and the head-in-pillow, can occur. These defects will be discussed later. Generally, a good compromise is a ramp rate of 1°C/second. This ramp rate also makes the reflow math pretty easy. For example, for a typical lead-free solder, the reflow temperature rises from 25°C (ambient) to 245°C (peak). This temperature change is a delta of 220°C. Because the suggested ramp rate is 1°C/second, the delta of 220°C represents 220 seconds, or 3 minutes 40 seconds, in the heating portion of the oven. The time is adjusted by the belt speed. For a 4 minute profile, if the total heating length of the oven is 120" you divide 120" by 4 minutes, which gives a belt speed of 30"/min. Of course, all this works well if the assembly is not very large and/or the board very thick. Below is some discussion on when to use soak profiles, but the target

is to get as close as possible to a RTP type profile.

Oven zone temperature is also important in optimizing the ramp rate. This optimization can be achieved by using the same delta between oven zones, for all zones in the oven, if possible—especially for all except the last two zones where the peak temperature and time-above-liquidus are controlled. For example, for many assemblies, one could very effectively use the following temperatures in a 7 zone oven: 110/135/160/185/205/230/255°C with a 25°C delta between each zone. If the peak temperature and/or time-above-liquidus are out of specification, the last two zones can be adjusted to meet the specification. The belt speed can also be adjusted, but care should be taken not to deviate too far from the recommended 1°C/s ramp rate.

It should be noted that starting the first zone at 100-130°C will help drive off solder paste solvents and other volatiles, minimizing slump, solder balls and splatter, and flux vapor condensation. Such condensation can clog the oven ventilation system.

Soak

The ramp-to-peak profile is not an answer to all applications. As mentioned previously, the RTP profile may not be appropriate for large assemblies, those with very large components, or assemblies with back planes. The inappropriateness of the RTP profile will be noted by a very large temperature delta between thermocouples at time-above-liquidus and peak temperature. Such assemblies require a soak or plateau where the delta between thermocouples is eliminated or greatly minimized. A high temperature soak just prior to reflow of the alloy will minimize the temperature delta. For example, for a SAC305 alloy, a soak temperature of 190-210°C is recommended. The length of the soak will be determined by the size of the assembly, or more importantly, the delta T between thermocouples. A larger delta T will require a longer soak. Unfortunately, this taxes the solder paste, especially for smaller paste deposits.

Another reason to use a soak profile is to reduce voiding in BGA or leadless components, such as QFNs (quad-flat pack no-leads). With the higher reflow temperatures associated with lead-free processing, and the higher surface tension of lead-free solders, such as SAC (tin-silver-copper), the soak profile isn't nearly as effective as for tin-lead assembly and may cause a number of other solder defects, such as graping or the head-in-pillow (HIP).

HIP/Graping

Graping Phenomenon

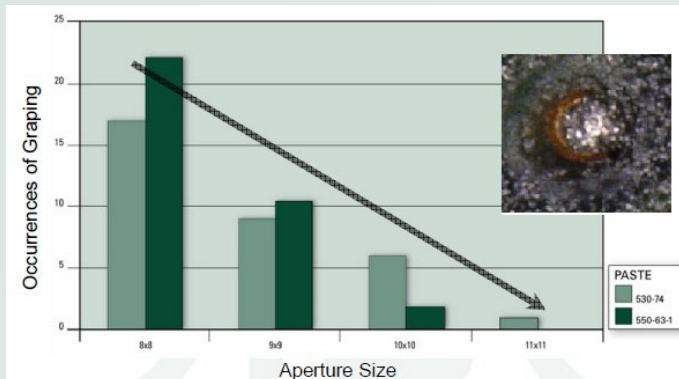


Figure 20. Occurrences of graping as aperture size decreases (from left to right).

Graping may, in some instances, be merely a cosmetic issue. However, as seen in Figure 20, the instance of graping increases as pad/aperture size and the amount of solder paste present decreases. Because there is so little solder present for these smaller components, the graping effect can easily result in a solder defect, as the mechanical and electrical connection is compromised.

In addition, the higher reflow temperature necessary to reflow most lead-free solders increases the graping phenomenon. During the heating process, the typical lead-free solder alloy does not become molten until 217°C, however the flux viscosity can decrease with every degree change in temperature. As

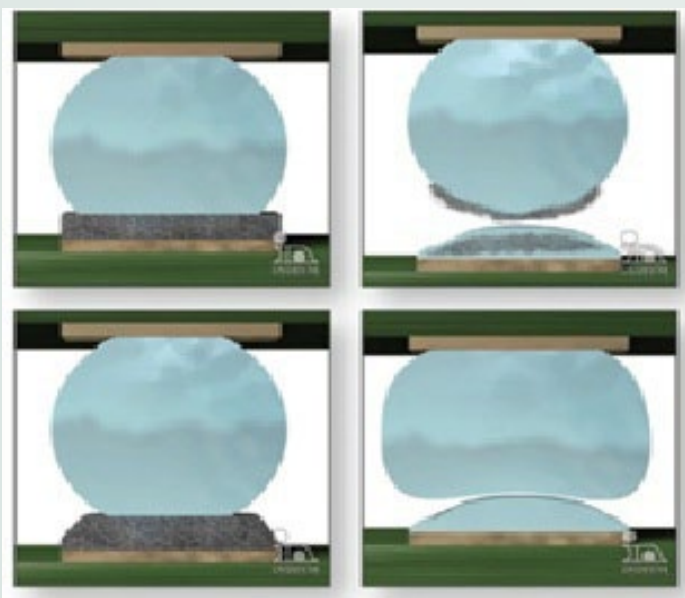


Figure 21. Progression of head-in-pillow defect.

the flux heats up, it begins to spread downward and outward, exposing the solder particles at the top of the solder paste deposit. With a lack of flux, the solder particles at the top of the solder paste deposit become highly oxidized. These oxides will inhibit the full coalescence of the particles into the solder joint. The unreflowed particles often exhibit the appearance of a cluster of grapes.

There are a number of causes for head-in-pillow defect; however, not all will be discussed here. A significant cause is related to BGA-type components where package warpage occurs during the reflow profile (Figure 21).

The packages can warp in different ways, but the end result is the same, open circuits. Separation of a solder sphere or spheres from the solder paste during reflow cause both the solder spheres and solder paste to oxidize and perhaps even to cool slightly before the warpage is relaxed, preventing the two from coalescing together. Instead, an impression is left in the surface, hence the term head-in-pillow, or head-on-pillow, as some refer to it.

Both head-in-pillow and graping are aggravated by longer and/or hotter reflow profiles and as the size of the solder paste deposit decreases.

Since both issues are seen more predominantly as the physical size of the solder paste deposit decreases, it is even more imperative that area ratio guidelines are followed and that the stencil printing process is set up to provide a good and consistent paste volume.

As for the reflow profile, anything that can be done to diminish the total heat excursion to the assembly will give better results. This can be accomplished by lowering the peak temperature, decreasing the time-above-liquidus, minimizing any soak, and/or decreasing the total time in the heating portion of the oven. Obviously enough heat needs to be provided to create the necessary intermetallic to create an electrical/mechanical connection. Generally, the peak temperature for lead-free soldering is 235-245°C and time-above-liquidus 30-90 seconds. Also note there is a balance between peak temperature and time-above-liquidus. The lower the peak temperature, the longer the time-above-liquidus needs to be to create a good bond. So, a peak temperature of 245°C and a time-above-liquidus of 90 seconds would certainly be on the high end, and may result in graping or head-in-pillow. On the other end, a peak temperature of 235°C and a time-above-liquidus of 30 seconds may result in a cold solder joint.

Summary

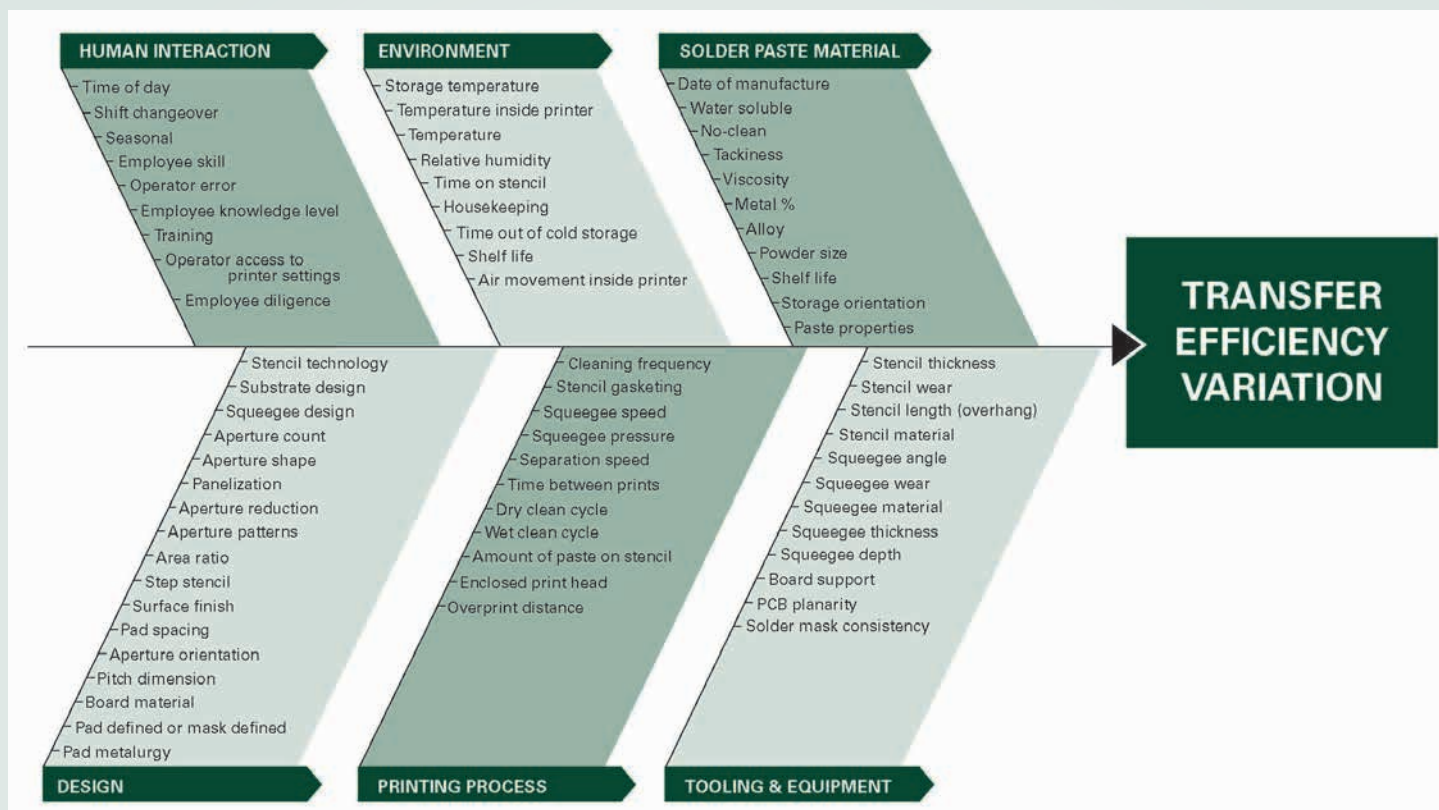
For today's technology, the higher reflow temperatures associated with the use of lead-free assembly processes and the continual movement towards miniaturization has caused a closer look at existing practices in stencil printing and reflow profiling. Even the practice of storage and handling of solder paste, general housekeeping of the stencil printer, and the peripheral utensils used need to be re-visited to ensure success. It is well accepted that 60-70% of solder defects occur at the stencil printing process. Metrics such as area ratio and transfer efficiency play a significant role in stencil printing consistency. Paste measurement equipment provides an excellent tool in understanding, perhaps overlooked deficiencies, in a stencil printing process and for making data driven changes.

The reflow profile process window has changed from the once default soak type profile to a thermally friendly, less flux exhausting heat excursion provided by a RTP type profile. A slow gradual ramp utilizing the same delta between zone temperatures for at least all but the last two zones helps to accomplish this. A ramp rate of 1°C/second is a good balance for reflow profiling to minimize solder defects caused by either too slow, or too fast a ramp rate

References

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First published at SMTA's International Conference of Soldering and Reliability in Toronto, CA, May 2012.



Sample Area Ratio Chart

Pad Size (mm)	0.05	0.10	0.15	01005	0.20	0.25	0201	0.30	0.35	0.40	0.45	0.50
Aperture Width (mil)	2.0	3.9	5.9	7 x 8	7.9	9.8	10 x 12	11.8	13.8	15.7	17.7	19.7
Stencil Thickness (5.0 mil)	0.10	0.20	0.30	0.37	0.39	0.49	0.55	0.59	0.69	0.79	0.89	0.98
Stencil Thickness (4.5 mil)	0.11	0.22	0.33	0.41	0.44	0.55	0.61	0.66	0.77	0.87	0.98	1.09
Stencil Thickness (4.0 mil)	0.12	0.25	0.37	0.47	0.49	0.62	0.68	0.74	0.86	0.98	1.11	1.23
Stencil Thickness (3.5 mil)	0.14	0.28	0.42	0.53	0.56	0.70	0.78	0.84	0.98	1.12	1.27	1.41
Stencil Thickness (3.0 mil)	0.16	0.33	0.49	0.62	0.66	0.82	0.91	0.98	1.15	1.31	1.48	1.64
Stencil Thickness (2.5 mil)	0.20	0.39	0.59	0.75	0.79	0.98	1.09	1.18	1.38	1.57	1.77	1.97